D 74380 (Pages : 4) Name.....

Reg. No.....

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FIRST SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2014

(CUCBCSS-U.G.)

Complementary Course—Mathematics

MAT IC 01—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer all **twelve** *questions.*

1. Evaluate $\lim_{x \to -5} \frac{x^2+3x-10}{x+5}$.

2. At what points are the function $y = \frac{\cos x}{x}$ continuous ?

3. Find the slope of $f(x) = x^2 + 1$ at (3, 7).

4. Find the derivative of $y = x^2$ using the definition of derivative.

5. Find the second derivative of $y = \frac{1}{3x^2} - \frac{5}{2x}$.

6. How fast does the area of a circle change with respect to the diameter when the diameter is 8 m ?

7. Find the critical points of $f(x) = \frac{2x^3 - 3x^2}{6}$.

8. Graph the parabola $y = x^2$.

9. Find $\lim_{x \to \infty} \frac{2x+3}{5x+7}$.

10. Evaluate the sum of the first 20 cubes.

11. State the mean value theorem for definite integrals.

12. Find the intersection points of $f(x) = 2 - x^2$ and $g(x) = -x$.

(12 × 1 = 12 marks)

Turn over

Section B

Answer all **nine** *questions.*

13. If $\sqrt{5-2x^2} \le f(x) \le \sqrt{5-x^2}, -1 \le x \le 1$, find $\underset{x \to 0}{\text{Lt}} f(x)$.

14. Prove that $\underset{x \to 1}{\text{Lt}} f(x) = 1$ if $f(x) = \begin{cases} x^2, x \neq 1 \\ 2, \ x = 1 \end{cases}$.

15. Suppose $\underset{x \to c}{\text{Lt}} f(x) = 5$ and $\underset{x \to c}{\text{Lt}} g(x) = -2$. Find :

    (i) $\underset{x \to c}{\text{Lt}} [f(x) + 3g(x)]$; and (ii) $\underset{x \to c}{\text{Lt}} \dfrac{f(x)}{f(x) - g(x)}$.

16. Find the derivative of $y = \dfrac{1}{(x^2-1)(x^2+x+1)}$.

17. Find the equation of the tangent to the curve $y = x^3 - 4x + 1$ at (2, 1).

18. Find $\underset{x \to 0}{\text{Lt}} \dfrac{8x^2}{\cos x - 1}$.

19. Find the linearization of $f(x) = x^3 - x$ at $x = 1$.

20. Graph the function $y = \dfrac{1}{2x+4}$.

21. Find the area of the region enclosed by $y = x^2 - 2$ and $y = 2$.

22. Find the function $f(x)$ whose derivative is $\sin x$ and whose graph passes through (0, 2).

23. Find the derivatives of all orders of $y = {}^{x^5}/_{120}$.

24. State both parts of the fundamental theorem of calculus.

(9 × 2 = 18 marks)

Section C

Answer any **six** *questions.*

25. Show that $y = \sin\left({}^{1}/_{x}\right)$ has no limit point as x approaches zero from either side. Also sketch the graph of this function.

26. Evaluate $\lim_{x \to 1} \frac{x-1}{\sqrt{x+3}-2}$.

27. The curve $y = ax^2 + bx + c$ passes through (1, 2) and is tangent to $y = x$ at the origin. Find a, b, c.

28. State and prove the product rule for derivatives. Use it to find the derivative of $y = (x^2+1)(x^3+3)$.

29. Find the intervals on which $f(x) = \frac{x^2-3}{x-2}$, $x \neq 2$ is increasing and decreasing. Identify local extrema if they exist.

30. Define average value of an integrable function over a closed interval. Find the average value of $f(x) = -3x^2 - 1$ on [0,1]. where in the given interval does $f(x)$ assume its average value.

31. Show that $\lim_{x \to 0+} (1+x)^{1/x} = e$.

32. An object is dropped from the top of a 100 m high tower. Its height above ground after 't' seconds is $(100 - 4.9\, t^2)$ m. How fast is it falling 2 seconds after it is dropped ?

33. Find the derivative $\frac{d}{dx} \int_0^{\sqrt{x}} \cos t \, dt$ by (i) evaluating the integral and differentiating the result ; and

(ii) by differentiating the integral directly.

(6 × 5 = 30 marks)

Section D

Answer any **two** *questions.*

34. (i) Find the area of the region enclosed by the curves $x + 4y^2 = 4$ and $x + y^4 = 1$ for $x \geq 0$.

(ii) Find the volume of the solid generated by revolving the region bounded by $y = x^2, y = 0, x = 2$ about the x-axis.

35. (i) Graph the function $y = x^4 - 4x^3 + 10$ by finding the first and second derivative.

(ii) Evaluate $\lim_{x \to 0} \frac{x^2}{\ln(\sec x)}$.

(iii) Evaluate $\sum_{k=1}^{4} \cos k\pi$.

36. (i) Let $f(x)=\begin{cases}3-x, x<2\\ \frac{x}{2}+1, x>2\end{cases}$. Find :

(a) $\underset{x\to 2+}{\text{Lt}} f(x)$ and $\underset{x\to 2-}{\text{Lt}} f(x)$.

(b) Does $\underset{x\to 2}{\text{Lt}} f(x)$ exist ? why or why not ?

(c) $\underset{x\to 4-}{\text{Lt}} f(x)$ and $\underset{x\to 4+}{\text{Lt}} f(x)$.

(d) Does $\underset{x\to 4}{\text{Lt}} f(x)$ exist ? Why or why not ?

(ii) Show that the line $y = mx + b$ is its own tangent at any point $(x_0, mx_0 + b)$.

(2 × 10 = 20 marks)